

Supplementary Material for HCPS: Heuristic Candidate-space Policy Search for Agile Earth Observation Satellite Scheduling

Satellite Orbital Parameters

Table S1 lists the Keplerian elements of the six satellites used in the experiments.

Table S1: Keplerian elements of satellite orbits.

Satellite	a (km)	e	i ($^{\circ}$)	Ω ($^{\circ}$)	ω ($^{\circ}$)	ν ($^{\circ}$)
sat1	7200	0.000627	96.576	175.72	0.075	0
sat2	7200	0.000627	96.576	145.72	30.075	0
sat3	7200	0.000627	96.576	115.72	60.075	0
sat4	7200	0.000627	96.576	85.72	90.075	0
sat5	7200	0.000627	96.576	55.72	120.075	0
sat6	7200	0.000627	96.576	25.72	150.075	0

Complete Performance Comparison

Table S2 presents the full results of target completion rate F_s and runtime for all five algorithms across all 15 problem scales (100 to 1500 targets, step 100). Values are reported as mean $\pm$ standard deviation over 20 independent test instances per scale. The best two results per row (based on mean values) are highlighted in bold.

Table S2: Full performance comparison (mean $\pm$ std over 20 test instances per scale).

Scale	Metric	A-ALNS	GRILS	GA	RL-GA	HCPS
100	F_s (%)	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0
	t (s)	0.09 $\pm$ 0.01	0.04 $\pm$ 0.01	0.12 $\pm$ 0.02	0.16 $\pm$ 0.02	0.51 $\pm$ 0.04
200	F_s (%)	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0
	t (s)	38.64 $\pm$ 3.10	0.08 $\pm$ 0.01	0.48 $\pm$ 0.03	0.51 $\pm$ 0.03	0.93 $\pm$ 0.07
300	F_s (%)	94.13 $\pm$ 0.56	100.0 $\pm$ 0.0	99.78 $\pm$ 0.34	99.75 $\pm$ 0.42	100.0 $\pm$ 0.0
	t (s)	100.17 $\pm$ 5.23	0.19 $\pm$ 0.03	12.20 $\pm$ 0.78	12.72 $\pm$ 0.85	2.05 $\pm$ 0.14
400	F_s (%)	87.24 $\pm$ 0.89	98.30 $\pm$ 0.85	95.74 $\pm$ 0.12	95.97 $\pm$ 0.88	100.0 $\pm$ 0.0
	t (s)	191.55 $\pm$ 9.12	18.66 $\pm$ 9.98	19.40 $\pm$ 1.02	21.38 $\pm$ 1.15	3.11 $\pm$ 0.11
500	F_s (%)	79.11 $\pm$ 1.03	96.27 $\pm$ 0.58	88.64 $\pm$ 0.72	89.21 $\pm$ 1.05	97.64 $\pm$ 0.91
	t (s)	298.17 $\pm$ 11.45	32.82 $\pm$ 2.71	27.92 $\pm$ 1.34	30.77 $\pm$ 1.48	4.17 $\pm$ 0.19
600	F_s (%)	74.11 $\pm$ 1.21	92.10 $\pm$ 1.17	81.50 $\pm$ 1.23	83.98 $\pm$ 1.17	93.28 $\pm$ 1.39
	t (s)	403.69 $\pm$ 14.23	50.10 $\pm$ 3.12	36.80 $\pm$ 1.67	40.29 $\pm$ 1.85	5.02 $\pm$ 0.22
700	F_s (%)	68.58 $\pm$ 1.34	87.04 $\pm$ 1.33	75.50 $\pm$ 1.34	76.31 $\pm$ 1.28	88.52 $\pm$ 1.25
	t (s)	593.19 $\pm$ 25.34	69.49 $\pm$ 2.91	46.30 $\pm$ 1.89	50.62 $\pm$ 2.01	6.81 $\pm$ 0.27
800	F_s (%)	63.33 $\pm$ 1.45	81.49 $\pm$ 1.65	70.61 $\pm$ 1.45	72.55 $\pm$ 1.39	83.17 $\pm$ 1.32
	t (s)	794.65 $\pm$ 27.76	89.81 $\pm$ 2.7	56.13 $\pm$ 2.03	60.01 $\pm$ 2.15	8.20 $\pm$ 0.31
900	F_s (%)	60.48 $\pm$ 1.56	77.98 $\pm$ 1.30	65.37 $\pm$ 1.53	71.62 $\pm$ 1.48	80.56 $\pm$ 1.48
	t (s)	1044.83 $\pm$ 30.89	112.88 $\pm$ 3.67	66.57 $\pm$ 2.24	86.49 $\pm$ 2.56	9.79 $\pm$ 0.36
1000	F_s (%)	56.69 $\pm$ 1.97	74.51 $\pm$ 1.01	61.38 $\pm$ 1.62	68.31 $\pm$ 1.56	77.39 $\pm$ 1.82
	t (s)	1312.90 $\pm$ 42.23	136.72 $\pm$ 4.23	77.32 $\pm$ 2.45	97.04 $\pm$ 2.89	11.23 $\pm$ 0.42
1100	F_s (%)	53.50 $\pm$ 1.78	71.60 $\pm$ 1.22	58.32 $\pm$ 1.31	62.90 $\pm$ 1.64	73.87 $\pm$ 1.78
	t (s)	1604.88 $\pm$ 51.67	162.30 $\pm$ 4.68	88.99 $\pm$ 2.67	109.79 $\pm$ 3.12	13.32 $\pm$ 0.48
1200	F_s (%)	51.22 $\pm$ 1.89	67.51 $\pm$ 1.04	54.98 $\pm$ 1.19	59.42 $\pm$ 1.72	70.21 $\pm$ 1.34
	t (s)	1897.11 $\pm$ 64.34	190.06 $\pm$ 5.34	100.60 $\pm$ 2.89	122.71 $\pm$ 3.45	15.85 $\pm$ 0.55
1300	F_s (%)	48.99 $\pm$ 1.01	64.73 $\pm$ 0.94	52.05 $\pm$ 1.17	56.27 $\pm$ 1.80	66.94 $\pm$ 1.21
	t (s)	2105.45 $\pm$ 72.56	219.14 $\pm$ 5.99	111.39 $\pm$ 3.12	131.08 $\pm$ 3.78	17.95 $\pm$ 0.62
1400	F_s (%)	46.58 $\pm$ 1.12	62.90 $\pm$ 1.01	49.54 $\pm$ 1.14	54.32 $\pm$ 1.87	63.38 $\pm$ 1.08
	t (s)	2395.31 $\pm$ 73.67	251.31 $\pm$ 7.45	124.09 $\pm$ 3.34	144.04 $\pm$ 4.12	20.62 $\pm$ 0.69
1500	F_s (%)	44.65 $\pm$ 0.93	60.80 $\pm$ 0.69	47.08 $\pm$ 1.21	51.81 $\pm$ 1.94	60.95 $\pm$ 0.85
	t (s)	2668.85 $\pm$ 80.78	282.62 $\pm$ 6.01	136.44 $\pm$ 3.56	149.70 $\pm$ 4.45	23.28 $\pm$ 0.76